

Def.

Given four complex numbers  $a, b, c, d \in \mathbb{C}$  with  $ad - bc \neq 0$ , a map

$$W = T(z) = \frac{az + b}{cz + d}$$

is called a linear fractional transformation.

Observation for the linear fraction.

The condition  $ad - bc \neq 0$  gives that the  $T(z)$  is non-constant, assume  $ad - bc = 0$ .

If  $c \neq 0$ , then

$$\frac{az + b}{cz + d} = \frac{\frac{a}{c} \cdot (cz + d) + b - \frac{ad}{c}}{cz + d} = \frac{a}{c}.$$

If  $a \neq 0$ , then

$$\frac{az + b}{cz + d} = \frac{az + b}{cz + \frac{bc}{a}} = \frac{a}{c} \cdot \left( \frac{z + \frac{b}{a}}{z + \frac{bc}{a^2}} \right) = \frac{a}{c} \quad \left( z \neq -\frac{b}{a} \right).$$

If  $a = 0$ , then either  $b = 0$  or  $c = 0$ .

If  $b = 0$ , then  $T \equiv 0$ .

If  $c = 0$ , then  $T \equiv \frac{b}{d}$ .

If  $c = 0$ , then either  $a = 0$  or  $d = 0$ .

If  $a = 0$ , then  $T \equiv \frac{b}{d}$ .

If  $d = 0$ , then  $T$  is not defined.

Now, we assume  $ad - bc \neq 0$ , then on  $\mathbb{C} \setminus \{-\frac{d}{c}\}$ .

$$W = T(z) = \begin{cases} \frac{a}{c} - \left[ \frac{ad - bc}{c^2} \right] \cdot \frac{1}{z + \frac{d}{c}} & \text{if } c \neq 0 \\ \frac{a}{d}z + \frac{b}{d} & \text{if } c = 0, \end{cases}$$

Therefore, the  $T$  is injective.

Thm.

Define  $\mathbb{H} \stackrel{\text{def}}{=} \{z \in \mathbb{C} \mid \text{Im } z > 0\}$ . (That is, the upper plane)

A map  $F: \mathbb{H} \rightarrow \mathbb{D}: z \mapsto \frac{i-z}{i+z}$  is Conformal map,

and  $G: \mathbb{D} \rightarrow \mathbb{H}: w \mapsto i \cdot \frac{1-w}{1+w}$  is the inverse of  $F$ .

Proof. Given  $a+ib \in \mathbb{H}$  ( $b > 0$ ),

$$\|(0,1) - (a,b)\|^2 = a^2 + (b-1)^2 < \|(0,1) - (a,b)\|^2 = a^2 + (b+1)^2.$$

Therefore for all  $z \in \mathbb{H}$ ,  $|F(z)| < 1$ , so the  $F$  is well-defined.

And, given  $u+iv \in \mathbb{D}$  ( $u^2+v^2 < 1$ ),

$$\begin{aligned} \text{Im}(G(w)) &= \text{Re} \left[ \frac{1-(u+iv)}{1+u+iv} \right] = \text{Re} \left[ \frac{(1-u-iv)(1+u-iv)}{(1+u)^2+v^2} \right] \\ &= \frac{1-u^2-v^2}{(1+u)^2+v^2} > 0 \end{aligned}$$

Therefore the  $G$  is also well-defined. And, these are rational of polynomials, so holomorphic. Now, given  $z \in \mathbb{H}$ ,

$$G(F(z)) = i \cdot \frac{1 - \frac{i-z}{i+z}}{1 + \frac{i-z}{i+z}} = i \cdot \frac{i+z - i+z}{i+z + i-z} = z, \text{ and similarly } F \circ G = \text{id}.$$

So, these are bijection with  $G = F^{-1}$   $\square$

Observe that for a real number  $x \in \mathbb{R}$ ,

$$F(x) = \frac{i-x}{i+x} = \frac{(i-x)/(-i+x)}{1+x^2} = \frac{1+2xi-x^2}{1+x^2} = \frac{1-x^2}{1+x^2} + i \cdot \frac{2x}{1+x^2},$$

and parametrize  $x = \tan t$  ( $-\frac{\pi}{2} < t < \frac{\pi}{2}$ ), then

$$\sin 2a = 2 \cdot \sin a \cdot \cos a = \frac{2 \tan a}{1 + \tan^2 a} \quad \text{and} \quad \cos 2a = \frac{1 - \tan^2 a}{1 + \tan^2 a}, \quad \text{so}$$

$$F(w) = \cos 2t + i \cdot \sin 2t = e^{2it}.$$